

Ejercicio 1 (estatico)

ID de base: 10.24.0.0/24

Area	Host	Red asignada	Mascara	Primera IP	Ultima IP	Broadcast
Admin	40	10.24.0.0	/26 -192	10.24.0.1	10.24.0.62	10.24.0.63
Ventas	30	10.24.0.64	/27 -224	.65	.94	.95
Soporte	14	10.24.0.96	/28 -240	.97	.110	.111
Gerencia	10	10.24.0.112	/28 -240	.113	.126	.127
Enlace R1-R2	2	10.24.0.128	/30 -252	.129	.130	.131
Enlace R2-R3	2	10.24.0.132	/30 -252	.133	.134	.135

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① $2^6 - 2 \geq 45$
 $64 - 2 \geq 45$
 $62 \geq 45$
 $32 - 6 = 26$
 $/26$
 $= 255.255.255.192$
 $256 - 192 = 64$

② $2^5 - 2 \geq 30$
 $32 - 2 \geq 30$
 $30 \geq 30$
 $32 - 5 = 27$
 $/27$
 $= 255.255.255.224$
 $256 - 224 = 32$

③ $2^4 - 2 \geq 14$
 $16 - 2 \geq 14$
 $14 \geq 14$
 $32 - 4 = 28$
 $/28$
 $= 255.255.255.240$
 $256 - 240 = 16$

④ $2^4 - 2 \geq 10$
 $16 - 2 \geq 10$
 $14 \geq 10$
 $32 - 4 = 28$
 $/28$
 $= .240$
 $= 16$

⑤ ⑥ $2^2 - 2 \geq 2$
 $4 - 2 \geq 2$
 $2 \geq 2$
 $32 - 2 = 30$
 $/30$
 $= .252$
 $256 - 252 = 4$

Tabla de direccionamiento

Tabla de direccionamiento				
Dispositivo	Interfaz	IP	Máscara	Gateway
PC1	NIC	10.24.0.2	.192	10.24.0.1
R1	Fa0/0	10.24.0.1		
PC2	NIC	10.24.0.66	.224	10.24.0.65
R1	Fa0/1	10.24.0.65		
PC3	NIC	10.24.0.98	.240	10.24.0.97
R2	Fa0/0	10.24.0.97		
PC4	NIC	10.24.0.114	.240	10.24.0.113
R3	Fa0/0	10.24.0.113		
R1	Fa1/0	10.24.0.129	.252	
R2	Fa0/1	. . .130		
R2	Fa1/0	. . .133	.252	
R3	Fa0/1	. . .134		

Ejercicio 2 (dinamico)

IP de base: 192.24.0.0/24						
Area	Host	Red asignada	Máscara	Primera IP	Ultima IP	Broadcast
PC1	40	192.24.0.0	.192	192.24.0.1	192.24.0.62	192.24.0.63
PC2	28	.64	.224	.65	.94	.95
PC3	18	.96	.224	.97	.126	.127
PC4	9	.128	.240	.129	.142	.143
PC5	5	.144	.248	.145	.150	.151
ER1-R2	2	.152	.252	.153	.154	.155
ER1-R3	2	.156	.252	.157	.158	.159
ER2-R5	2	.160	.252	.161	.162	.163
R3-R4	2	.164	.252	.165	.166	.167
R4-R5	2	.168	.252	.169	.170	.171

$2^0 - 2 \geq 40$ $64 - 2 \geq 40$ $62 \geq 40$ $32 - 6 = 26$ 126 $.192$ $256 - 192 = 64$	$2^5 - 2 \geq 28$ $32 - 2 \geq 28$ $30 \geq 28$ $32 - 5 = 27$ 127 $.224$ $256 - 224 = 32$	$2^5 - 2 \geq 18$ $32 - 2 \geq 18$ $30 \geq 18$ 127 $.240$ $\text{salto} = 32$	$2^4 - 2 \geq 9$ $16 - 2 \geq 9$ $14 \geq 9$ $32 - 4 = 28$ 128 $.240$ $256 - 240 = 16$	$2^3 - 2 \geq 5$ $8 - 2 \geq 5$ $6 \geq 5$ $32 - 3 = 29$ 129 $.248$ $256 - 248 = 8$	$2^2 - 2 \geq 2$ $4 - 2 \geq 2$ $2 \geq 2$ $32 - 2 = 30$ 130 $.252$ $256 - 252 = 4$
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Tabla de direccionamiento

Tabla de direccionamiento:

Dispositivo	IP	Interfaz	Mascara	Gateway
PC1	192.24.0.2	NIC	.192	192.24.0.1
✓ R5	192.24.0.1	Fa0/0		
PC2	.66	NIC	.224	192.24.0.65
✓ R2	192.24.0.65	Fa0/0		
PC3	.98	NIC	.224	192.24.0.97
✓ R3	192.24.0.97	Fa0/0		
PC4	.130	NIC	.240	192.24.0.129
✓ R1	192.24.0.129	Fa0/0		
PC5	.146	NIC	.248	192.24.0.145
✓ R4	192.24.0.145	Fa0/0		PC5, R3, R5
✓ R1	.153	Fa0/1	.252	
✓ R2	.154	Fa0/1		
✓ R1	.157	Fa1/0	.252	
✓ R3	.158	Fa0/1		
✓ R2	.161	Fa1/0	.252	
✓ R5	.162	Fa0/1		
✓ R3	.165	Fa1/0	.252	
✓ R4	.166	Fa0/1		
✓ R4	.169	Fa1/0	.251	
✓ R5	.170	Fa1/0		